

[EECS 304] Control I

Sheet (9) SSR Canonical Forms

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Fall 2026

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- Controllable Canonical Form
- Observable Canonical Form
- Diagonal Canonical Form
- How to check for Controllability and Observability

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Getting the SSR from the TF

- There are three forms of SSR:
 - ▶ Controllable Canonical Form
 - ▶ Observable Canonical Form
 - ▶ Diagonal Canonical Form

$$T.F. = \frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_1s + a_0}$$

Note: s^n in the T.F. must have a coefficient of 1, if not, divide the whole T.F. by its coefficient.

Controllable Canonical Form

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_1s + a_0}$$

$$\dot{\underline{x}} = A\underline{x} + B\underline{u}$$

$$y = C\underline{x} + D\underline{u}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dots \\ 1 \end{bmatrix}, \quad C = [b_0 \quad b_1 \quad \dots \quad b_{n-1}]$$

Observable Canonical Form

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_1s + a_0}$$

$$\dot{\underline{x}} = A\underline{x} + B\underline{u}$$

$$y = C\underline{x} + D\underline{u}$$

$$A = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} b_0 \\ b_1 \\ \dots \\ b_{n-1} \end{bmatrix}, \quad C = [0 \quad 0 \quad \dots \quad 1]$$

- Relation between observable and controllable canonical forms:

- ▶ $A_{\text{observable}} = A_{\text{controllable}}^T$
- ▶ $B_{\text{observable}} = C_{\text{controllable}}^T$
- ▶ $C_{\text{observable}} = B_{\text{controllable}}^T$

Diagonal Canonical Form

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_1s + a_0}$$

$$\dot{\underline{x}} = A\underline{x} + B\underline{u}$$

$$y = C\underline{x} + D\underline{u}$$

Transform the T.F. to partial fractions: $\frac{Y(S)}{U(S)} = \frac{K_1}{S + P_1} + \frac{K_2}{S + P_2} + \dots + \frac{K_n}{S + P_n}$

$$A = \begin{bmatrix} -P_1 & 0 & \dots & \dots & 0 \\ 0 & -P_2 & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & -P_n \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \dots \\ 1 \end{bmatrix}, \quad C = [K_1 \quad K_2 \quad \dots \quad K_n]$$

How to check for Controllability and Observability

Controllability:

- Get controllability matrix Q_c
- If $|Q_c| \neq 0$ or $\text{rank}(Q_c)=n$, this system is completely controllable.
- If $\text{rank}(Q_c)<n$, this system is not completely controllable.

$$Q_c = [B|AB|A^2B| \dots |A^{n-1}B]$$

- In **diagonal** systems, if the **B matrix** has a **complete row of zeroes**, the system will be **uncontrollable**.

How to check for Controllability and Observability

Observability:

- Get observability matrix Q_o
- If $|Q_o| \neq 0$ or $\text{rank}(Q_o)=n$, this system is completely observable.
- If $\text{rank}(Q_c)<n$, this system is not completely observable.

$$Q_c = \begin{bmatrix} C \\ CA \\ \dots \\ CA^{n-1} \end{bmatrix}$$

- In **diagonal** systems, if the **C matrix** has a **complete column of zeroes**, the system will be **unobservable**.

Problem 1

Problem 1

Transform the following transfer function into the observable and controllable canonical forms:

$$\frac{Y(s)}{R(s)} = \frac{4s^2 + 3s + 5}{s^3 + 2s^2 + 4s + 1}$$

Problem 1

Solution

Controllable Canonical Form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 5 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Observable Canonical Form:

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -4 \\ 0 & 1 & -2 \end{bmatrix} \underline{x} + \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \underline{x}$$

Problem 2

Problem 2

Transform the following transfer function into the diagonal form:

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 2s + 1}{(s + 1)(s + 0.5)(s + 2)}$$

Problem 2

Solution

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 2s + 1}{(s + 1)(s + 0.5)(s + 2)} = \frac{0}{s + 1} + \frac{1/3}{s + 0.5} + \frac{2/3}{s + 2}$$

Diagonal Canonical Form:

$$\dot{\underline{x}} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -0.5 & 0 \\ 0 & 0 & -2 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1/3 & 2/3 \end{bmatrix} \underline{x}$$

Problem 3

Problem 3

Given the following state space model:

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -21 & -16 & -6 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u, \quad y = [10 \quad 15 \quad 5] \underline{x}$$

- Test this model for both controllability and observability.
- Obtain the transfer function for this system.

Problem 3

Solution

a) To test the controllability, we must form the controllability matrix

$$Q_c = [B|AB|A^2B]$$

$$AB = \begin{bmatrix} 0 \\ 1 \\ -6 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ -21 & -16 & -6 \\ 126 & 75 & 20 \end{bmatrix}$$

$$Q_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -6 \\ 1 & -6 & 20 \end{bmatrix} \rightarrow |Q_c| \neq 0 \therefore \text{Rank}(Q_c) = 3$$

$\therefore$ System is controllable.

Problem 3

Solution

a) To test the observability, we must form the observability matrix

$$Q_o = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ -21 & -16 & -6 \\ 126 & 75 & 20 \end{bmatrix}$$

$$Q_o = \begin{bmatrix} 10 & 15 & 5 \\ -105 & -70 & -15 \\ 315 & 135 & 20 \end{bmatrix} \rightarrow |Q_o| \neq 0 \therefore \text{Rank}(Q_o) = 3$$

$\therefore$ System is observable.

Problem 3

Solution

b)

$$\therefore T.F. = C(sI - A)^{-1}B + D$$

$$\therefore (sI - A) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -21 & -16 & -6 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 21 & 16 & s+6 \end{bmatrix}$$

$$\therefore (sI - A)^{-1} = \frac{1}{s^3 + 6s^2 + 16s + 21} \begin{bmatrix} s^2 + 6s + 16 & s + 6 & 1 \\ -21 & s^2 + 6s & s \\ -21s & -16s - 21 & s^2 \end{bmatrix}$$

$$\therefore (sI - A)^{-1}B = \frac{1}{s^3 + 6s^2 + 16s + 21} \begin{bmatrix} 1 \\ s \\ s^2 \end{bmatrix}$$

$$\therefore C(sI - A)^{-1}B = \frac{5s^2 + 15s + 10}{s^3 + 6s^2 + 16s + 21}$$

Problem 3

Solution

Note:

We could have get the T.F. directly from the state equation as it is in the controllable canonical form.

$$\therefore T.F. = \frac{5s^2 + 15s + 10}{s^3 + 6s^2 + 16s + 21}$$

Problem 4

Problem 4

A system is described by:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Test this model for both controllability and observability.

Problem 4

Solution

→ We can test controllability directly from B matrix
∴ B matrix has a complete row of zeroes.
∴ The system is un-controllable.

→ Observability is studied by inspecting C matrix
∴ All columns of C are non-zero.
∴ The system is observable.

Thank you!

Questions?

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